MANIPAL UNIVERSITY

School of Computer Science and Engineering

Department of Computer Science and Engineering

AI-Based Crowd Monitoring and Safety Alert System

Submitted By:
Aditya Raj Jain
2427030117

Supervised By:
Dr. Mohit Kushwaha

Problem Statement

- Manual crowd monitoring is inefficient
- No real-time density analysis
- High risk in public spaces (events, stations)

Need of automated intelligent system

Objective

- Real-time person detection
- Crowd density analysis
- Zone-based overcrowding detection
- Alert generation system

Approach Overview

Video Analysis Process for Real-Time Monitoring

Input Video → YOLO Detection → Person Counting → Grid Mapping → Density Classification → Alert Generation → Output Display

Tech Stack

- Python
- OpenCV
- YOLOv8 (Ultralytics)
- NumPy

Methodology

- Detect people using YOLOv8
- Extract bounding boxes
- Map people to grid cells
- Count per cell
- Apply threshold logic

HOG vs YOLO

HOG:

- Traditional method
- Slow and less accurate

YOLO:

- Deep learning-based
- Faster and highly accurate
- Real-time performance

Grid-Based Density

- Frame divided into grid (4×4)
- Each cell counts number of people
- Zones classified:

  Green → Safe

  Yellow → Moderate

  Red → High density

Results

- Real-time detection achieved
- ~30–45 FPS performance
- Accurate detection using YOLO
- Effective zone-based alerts

Conclusion

- YOLO significantly improves accuracy
- Real-time monitoring achieved
- System scalable for real-world use

Future Work

- Multi-camera system
- Predictive analytics
- Smart city integration
- Edge deployment

REFERENCES

[1] Redmon, J., Divvala, S., Girshick, R., Farhadi, A.

"You Only Look Once: Unified, Real-Time Object Detection," IEEE, 2016.

[2] Bochkovskiy, A., Wang, C.Y., Liao, H.Y.M.

"YOLOv4: Optimal Speed and Accuracy of Object Detection," 2020.

[3] Jocher, G. et al.

"Ultralytics YOLOv8 Documentation," https://docs.ultralytics.com

[4] Dalal, N., Triggs, B.

"Histograms of Oriented Gradients for Human Detection," CVPR, 2005.

Thank You